Reconstructing Al6063 diffraction profiles from archived TIFF plots

Abstract

Five archived Al6063 X-ray diffraction profiles were digitized from TIFF plots. Four tracing methods were evaluated against the original solution-treated (ST) scan. The skeleton-based method reduced the pointwise intensity RMSE from 48.884 to 9.244 counts and reproduced the grid positions of three principal maxima. The four rolled and annealed profiles could be checked against their TIFFs. Their numerical measurement errors remain unknown.

Source data and calibration

The 2015 archive contains five 3390 × 2379 pixel TIFF plots and one Bruker RAW4.00 ASCII export. The ST export contains 938 readings from 35.00° to 81.85° in 0.05° 2θ steps. The header lists Cu $K\alpha_1$ at 1.5406 Å. The TIFFs provide the source traces. The export supplies the numerical comparison.

The batch method maps horizontal pixels 606 and 2918 to 30° and 90° 2θ. Each image has its own two-point intensity calibration. The algorithm thresholds the cropped grayscale plot, skeletonizes the stroke, fills gaps by interpolation, and applies a local correction at prominent tips before resampling at 0.05°.

ST benchmark across 938 matched angles

Tracing method	MAE	RMSE	Pearson r	Sum error
V1 Upper ink edge	8.758	48.884	0.98926	+16.52%
V2 Column median	6.622	29.693	0.99209	-3.77%
V3 Median with tip correction	6.545	29.930	0.99191	-2.71%
V4 Skeleton, new calibration and tip correction	3.162	9.244	0.99924	+0.61%

MAE and RMSE use the pointwise intensity residual in counts. Sum error is the signed difference in summed intensity divided by the reference sum. The ST scan informed method selection and is an in-sample benchmark. V3 recovered peak height at 65.05° but slightly increased scan-wide RMSE relative to V2.

Solution-treated reference scan

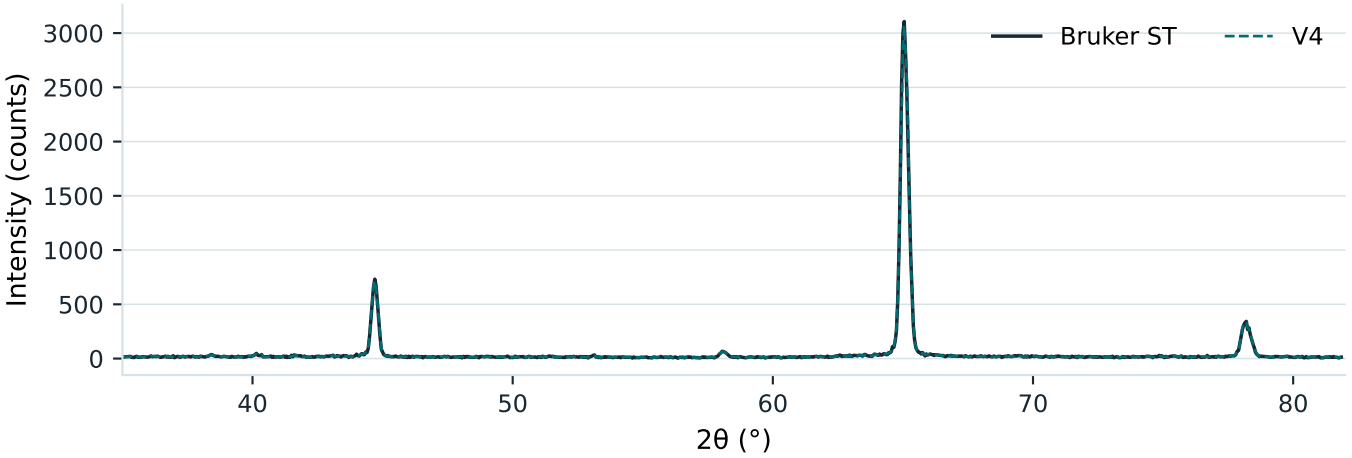


Figure 1. Original ST scan overlaid with the V4 reconstruction.

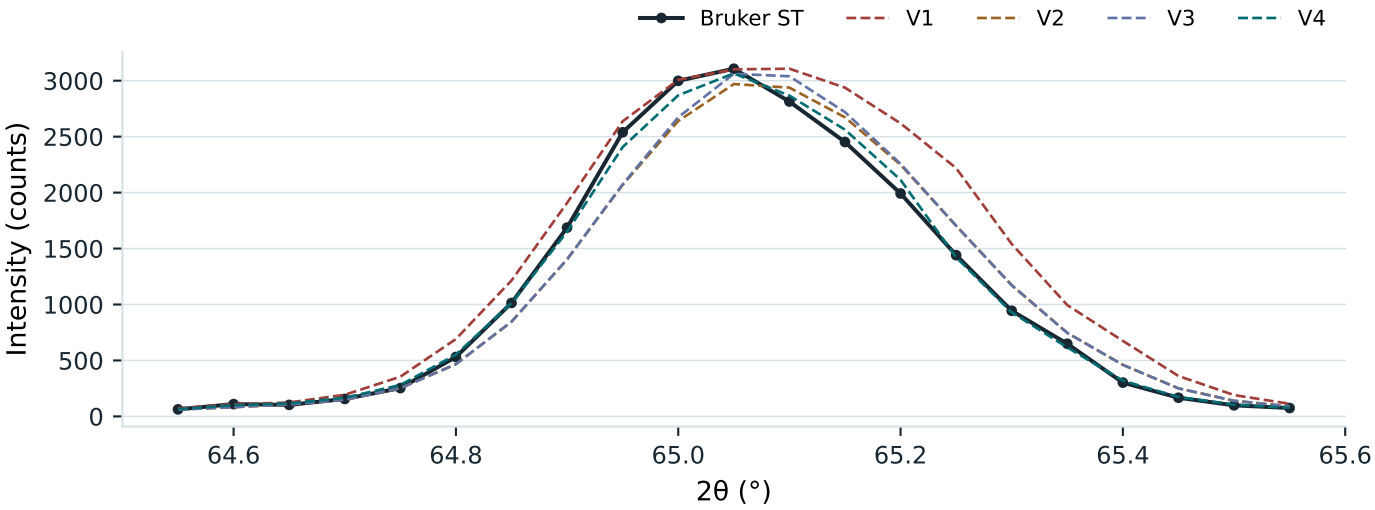


Figure 2. Detail of the (220) reflection for all four tracing methods.

Al reflection	Raw 2θ	V4 2θ	Height error	Width error	Area error
(200)	44.70°	44.70°	-2.51%	+1.42%	-0.09%
(220)	65.05°	65.05°	-1.35%	+0.37%	-0.80%
(311)	78.20°	78.20°	-2.65%	+0.33%	+0.57%

Al reflection assignments are tentative. Positions are grid maxima and widths are measured at half prominence. Areas use fixed windows and a linear background estimated from the median of three points at either end. V4 reproduced all three grid maxima, but the weak feature near 58° shifted by 0.05° and its area was 8.88% higher.

Five reconstructed profiles

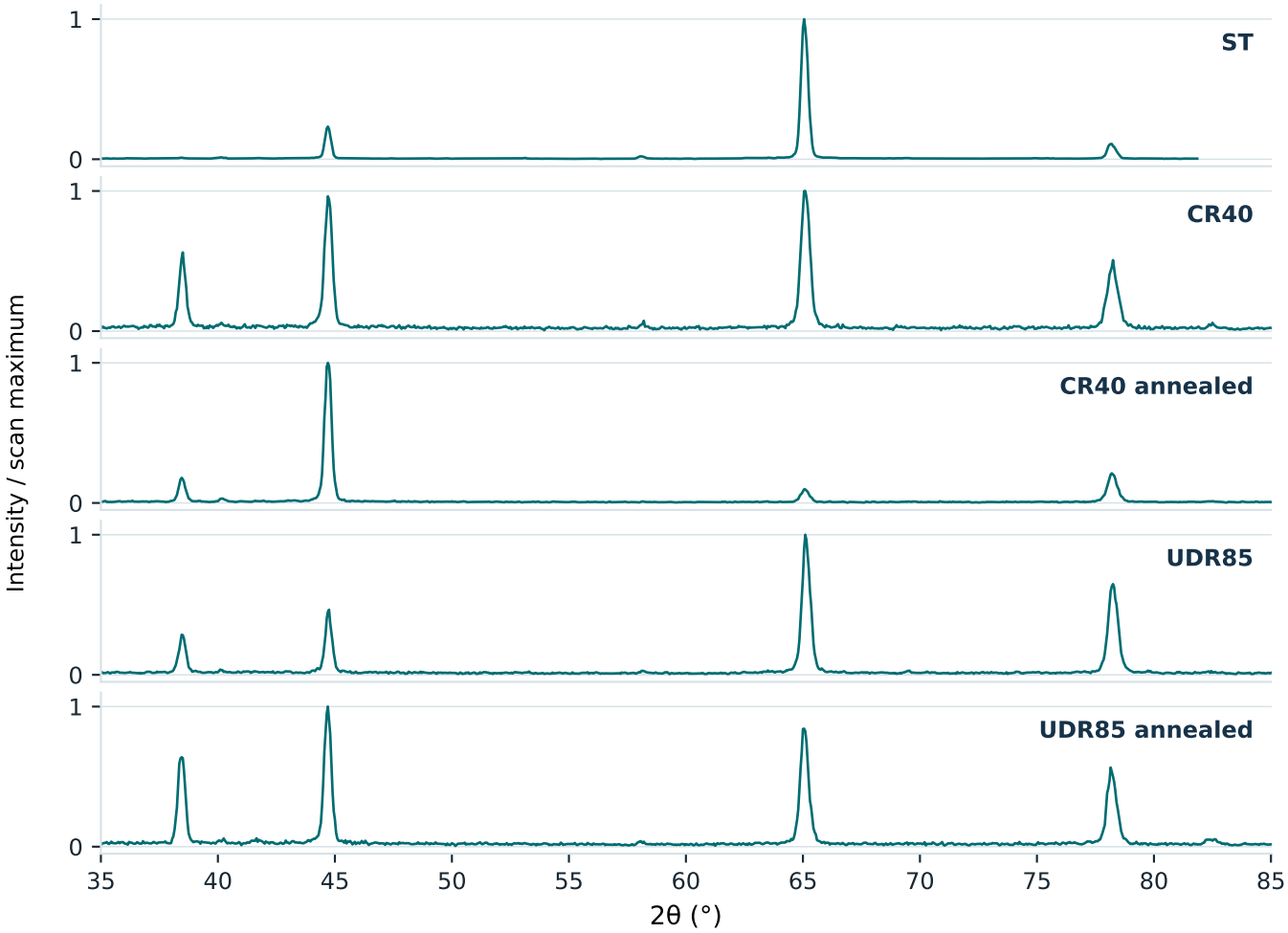


Figure 3. V4 profiles scaled to each scan's own maximum. The ST grid ends at 81.85°. The other four span 35.00° to 85.00° and contain 1,001 interpolated values each.

Image agreement and scope

The five batch CSVs reproduced the archived CSVs byte for byte. Mapping their points back to the source images flagged one UDR85 point at 78.25°, 21.10 pixels from the nearest dark stroke. The other samples had no points more than three pixels from the nearest dark stroke.

The plotted profiles support comparison of prominent peak positions and shapes. ST was the only numerical reference. The 0.05° export interval for the four other plots gives no independent information about their acquisition step. Quantitative texture, microstrain and crystallite size require additional acquisition or calibration data.